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Exfiltration Over C2 Channel

Adversaries may steal data by exfiltrating it over an existing command and control channel. Stolen data is encoded into the normal communications channel using the same protocol as command and control communications.

ID: T1041

Sub-techniques: No sub-techniques

Tactic: Exfiltration

Platforms: ESXi, Linux, Windows, macOS

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Procedure Examples

ID	Name	Description
S0045	ADVSTORESHELL	ADVSTORESHELL exfiltrates data over the same channel used for C2. ^[1]
G1030	Agrius	Agrius exfiltrated staged data using tools such as Putty and WinSCP, communicating with command and control servers. ^[2]
S1025	Amadey	Amadey has sent victim data to its C2 servers. ^[3]
S0584	AppleJeus	AppleJeus has exfiltrated collected host information to a C2 server. ^[4]
S0622	AppleSeed	AppleSeed can exfiltrate files via the C2 channel. ^[5]
G0022	APT3	APT3 has a tool that exfiltrates data over the C2 channel. ^[6]
G0050	APT32	APT32's backdoor has exfiltrated data using the already opened channel with its C&C server. ^[7]
G0087	APT39	APT39 has exfiltrated stolen victim data through C2 communications. ^[8]
C0046	ArcaneDoor	ArcaneDoor included use of existing command and control channels for data exfiltration. ^{[9][10]}
S9031	AshTag	AshTag has exfiltrated reconnaissance data on targeted systems to C2 servers. ^[11]
S0373	Astaroth	Astaroth exfiltrates collected information from its r1.log file to the external C2 server. ^[12]
S0438	Attor	Attor has exfiltrated data over the C2 channel. ^[13]
S1029	AuTo Stealer	AuTo Stealer can exfiltrate data over actor-controlled C2 servers via HTTP or TCP. ^[14]
S0031	BACKSPACE	Adversaries can direct BACKSPACE to upload files to the C2 Server. ^[15]
S1081	BADHATCH	BADHATCH can exfiltrate data over the C2 channel. ^{[16][17]}
S0234	Bandook	Bandook can upload files from a victim's machine over the C2 channel. ^[18]
S0239	Bankshot	Bankshot exfiltrates data over its C2 channel. ^[19]
S1246	BeaverTail	BeaverTail has exfiltrated data collected from victim devices to C2 servers. ^{[20][21][22]}
S0268	Bisonal	Bisonal has added the exfiltrated data to the URL over the C2 channel. ^[23]
G1043	BlackByte	BlackByte transmitted collected victim host information via HTTP POST to command and control infrastructure. ^[24]
S0520	BLINDINGCAN	BLINDINGCAN has sent user and system information to a C2 server via HTTP POST requests. ^{[25][26]}
S0657	BLUELIGHT	BLUELIGHT has exfiltrated data over its C2 channel. ^[27]
S0651	BoxCaon	BoxCaon uploads files and data from a compromised host over the existing C2 channel. ^[28]
S9015	BRICKSTORM	BRICKSTORM has uploaded files from the victim system to C2 servers. ^{[29][30][31][32][33][34][35]}
S1039	Bumblebee	Bumblebee can send collected data in JSON format to C2. ^[36]
C0017	C0017	During C0017, APT41 used its Cloudflare services C2 channels for data exfiltration. ^[37]
S0077	CallMe	CallMe exfiltrates data to its C2 server over the same protocol as C2 communications. ^[38]
S0351	Cannon	Cannon exfiltrates collected data over email via SMTP/S and POP3/S C2 channels. ^[39]
S0484	Carberp	Carberp has exfiltrated data via HTTP to already established C2 servers. ^{[40][41]}
S0572	Caterpillar WebShell	Caterpillar WebShell can upload files over the C2 channel. ^[42]
S0674	CharmPower	CharmPower can exfiltrate gathered data to a hardcoded C2 URL via HTTP POST. ^[43]
G0114	Chimera	Chimera has used Cobalt Strike C2 beacons for data exfiltration. ^[44]
S1149	CHIMNEYSWEEP	CHIMNEYSWEEP can upload collected files to the command-and-control server. ^[45]
S0667	Chrommme	Chrommme can exfiltrate collected data via C2. ^[46]
G0142	Confucius	Confucius has exfiltrated stolen files to its C2 server. ^[47]
G1052	Contagious Interview	Contagious Interview has exfiltrated data from a compromised host to actor-controlled C2 servers. ^{[48][49][50][51][20][52][53][54][21][22]}
S1024	CreepySnail	CreepySnail can connect to C2 for data exfiltration. ^[55]
S0115	Crimson	Crimson can exfiltrate stolen information over its C2. ^[56]

S0538	Crutch	Crutch can exfiltrate data over the primary C2 channel (Dropbox HTTP API). ^[57]
S1153	Cuckoo Stealer	Cuckoo Stealer can send information about the targeted system to C2 including captured passwords, OS build, hostname, and username. ^[58]
G1012	CURIUM	CURIUM has used IMAP and SMTPS for exfiltration via tools such as IMAPLoader. ^[59]
S0687	Cyclops Blink	Cyclops Blink has the ability to upload exfiltrated files to a C2 server. ^[60]
S1111	DarkGate	DarkGate uses existing command and control channels to retrieve captured cryptocurrency wallet credentials. ^[61]
S1021	DnsSystem	DnsSystem can exfiltrate collected data to its C2 server. ^[62]
S0600	Doki	Doki has used Ngrok to establish C2 and exfiltrate data. ^[63]
S0502	Drovorub	Drovorub can exfiltrate files over C2 infrastructure. ^[64]
S1159	DUSTTRAP	DUSTTRAP can exfiltrate collected data over C2 channels. ^[65]
S0062	DustySky	DustySky has exfiltrated data to the C2 server. ^[66]
S0024	Dyre	Dyre has the ability to send information staged on a compromised host externally to C2. ^[67]
S0377	Ebury	Ebury exfiltrates a list of outbound and inbound SSH sessions using OpenSSH's <code>known_hosts</code> files and <code>utmp</code> records. Ebury can exfiltrate SSH credentials through custom DNS queries or use the command <code>xsas</code> to send the process's ssh session's credentials to the C2 server. ^{[68][69]}
S0367	Emotet	Emotet has exfiltrated data over its C2 channel. ^{[70][71]}
S0363	Empire	Empire can send data gathered from a target through the command and control channel. ^{[72][73]}
S0568	EVILNUM	EVILNUM can upload files over the C2 channel from the infected host. ^[74]
S0696	Flagpro	Flagpro has exfiltrated data to the C2 server. ^[75]
S0381	FlawedAmmyy	FlawedAmmyy has sent data collected from a compromised host to its C2 servers. ^[76]
S0661	FoggyWeb	FoggyWeb can remotely exfiltrate sensitive information from a compromised AD FS server. ^[77]
C0001	Frankenstein	During <i>Frankenstein</i> , the threat actors collected information via <i>Empire</i> , which sent the data back to the adversary's C2. ^[73]
S1044	FunnyDream	FunnyDream can execute commands, including gathering user information, and send the results to C2. ^[78]
G0093	GALLIUM	GALLIUM used Web shells and HTRAN for C2 and to exfiltrate data. ^[79]
G0047	Gamaredon Group	A Gamaredon Group file stealer can transfer collected files to a hardcoded C2 server. ^{[80][81][82]}
S0493	GoldenSpy	GoldenSpy has exfiltrated host environment information to an external C2 domain via port 9006. ^[83]
S0588	GoldMax	GoldMax can exfiltrate files over the existing C2 channel. ^{[84][85]}
S0477	Goopy	Goopy has the ability to exfiltrate data over the Microsoft Outlook C2 channel. ^[86]
S0531	Grandoreiro	Grandoreiro can send data it retrieves to the C2 server. ^[87]
S0632	GrimAgent	GrimAgent has sent data related to a compromise host over its C2 channel. ^[88]
S0391	HAWKBALL	HAWKBALL has sent system information and files over the C2 channel. ^[89]
S1249	HexEval Loader	HexEval Loader has exfiltrated victim data using HTTPS POST requests to its C2 servers. ^{[51][52]}
G0126	Higaisa	Higaisa exfiltrated data over its C2 channel. ^[90]
C0038	HomeLand Justice	During <i>HomeLand Justice</i> , threat actors used HTTP to transfer data from compromised Exchange servers. ^[91]
S0376	HOPLIGHT	HOPLIGHT has used its C2 channel to exfiltrate data. ^[92]
S0431	HotCroissant	HotCroissant has the ability to download files from the infected host to the command and control (C2) server. ^[93]
S9007	HTTPTroy	HTTPTroy has exfiltrated encrypted data over the C2 channel using the <code>up-<FILENAME></code> command. ^[94]
S1022	IceApple	IceApple's Multi File Exfiltrator module can exfiltrate multiple files from a compromised host as an HTTP response over C2. ^[95]
S0434	Imminent Monitor	Imminent Monitor has uploaded a file containing debugger logs, network information and system information to the C2. ^[96]
S0604	Industroyer	Industroyer sends information about hardware profiles and previously-received commands back to the C2 server in a POST-request. ^[97]
S1245	InvisibleFerret	InvisibleFerret has used HTTP communications to the <code>/Uploads</code> URI for file exfiltration. ^[98]
S1132	IPsec Helper	IPsec Helper exfiltrates specific files through its command and control framework. ^[99]
G0004	Ke3chang	Ke3chang transferred compressed and encrypted RAR files containing exfiltration through the established backdoor command and control channel during operations. ^[100]
S0487	Kessel	Kessel has exfiltrated information gathered from the infected system to the C2 server. ^[101]
S1020	Kevin	Kevin can send data from the victim host through a DNS C2 channel. ^[102]
S0526	KGH_SPY	KGH_SPY can exfiltrate collected information from the host to the C2 server. ^[103]
G0094	Kimsuky	Kimsuky has exfiltrated data over its C2 channel. ^{[104][105]}
S0356	KONNI	KONNI has sent data and files to its C2 server. ^{[106][107][108]}
S1075	KOPILUWAK	KOPILUWAK has exfiltrated collected data to its C2 via POST requests. ^[109]
S9035	LAMEHUG	LAMEHUG can exfiltrate collected system information and documents to C2. ^{[110][111]}
S1160	Latrodectus	Latrodectus can exfiltrate encrypted system information to the C2 server. ^{[112][113]}
G0032	Lazarus Group	Lazarus Group has exfiltrated data and files over a C2 channel through its various tools and malware. ^{[114][115][116]}
G0065	Leviathan	Leviathan has exfiltrated data over its C2 channel. ^[117]

C0049	Leviathan Australian Intrusions	Leviathan exfiltrated collected data over existing command and control channels during Leviathan Australian Intrusions. ^[118]
S0395	LightNeuron	LightNeuron exfiltrates data over its email C2 channel. ^[119]
S1185	LightSpy	To exfiltrate data, LightSpy configures each module to send an obfuscated JSON blob to hardcoded URL endpoints or paths aligned to the module name. ^[120]
S1186	Line Dancer	Line Dancer exfiltrates collected data via command and control channels. ^[9]
S1188	Line Runner	Line Runner utilizes HTTP to retrieve and exfiltrate information staged using Line Dancer. ^[9]
S0680	LitePower	LitePower can send collected data, including screenshots, over its C2 channel. ^[121]
S9020	LODEINFO	LODEINFO can exfiltrate collected credentials and browser cookies to the C2 server. ^[122]
S0447	Lokibot	Lokibot has the ability to initiate contact with command and control (C2) to exfiltrate stolen data. ^[123]
G1014	LuminousMoth	LuminousMoth has used malware that exfiltrates stolen data to its C2 server. ^[124]
S1213	Lumma Stealer	Lumma Stealer has exfiltrated collected data over existing HTTP and HTTPS C2 channels. ^{[125][126]}
S1142	LunarMail	LunarMail can use email image attachments with embedded data for receiving C2 commands and data exfiltration. ^[127]
S0409	Machete	Machete's collected data is exfiltrated over the same channel used for C2. ^[128]
S1016	MacMa	MacMa exfiltrates data from a supplied path over its C2 channel. ^[129]
S1060	Mafalda	Mafalda can send network system data and files to its C2 server. ^[130]
S1182	MagicRAT	MagicRAT exfiltrates data via HTTP over existing command and control channels. ^[131]
S1169	Mango	Mango can use its HTTP C2 channel for exfiltration. ^[132]
S1156	Manjusaka	Manjusaka data exfiltration takes place over HTTP channels. ^[133]
S0652	MarkIRAT	MarkIRAT can exfiltrate locally stored data via its C2. ^[134]
S0459	MechaFlounder	MechaFlounder has the ability to send the compromised user's account name and hostname within a URL to C2. ^[135]
S1059	metaMain	metaMain can upload collected files and data to its C2 server. ^[136]
S0455	Metamorfo	Metamorfo can send the data it collects to the C2 server. ^[137]
S0084	Mis-Type	Mis-Type has transmitted collected files and data to its C2 server. ^[138]
S0083	Misdat	Misdat has uploaded files and data to its C2 servers. ^[138]
S1122	Mispadu	Mispadu can sends the collected financial data to the C2 server. ^{[139][140]}
S0079	MobileOrder	MobileOrder exfiltrates data to its C2 server over the same protocol as C2 communications. ^[38]
S1026	Mongall	Mongall can upload files and information from a compromised host to its C2 server. ^[141]
S9032	MuddyViper	MuddyViper has uploaded files to the C2 server. Additionally, MuddyViper has the ability to upload the specified file in chunks with sleep time between each chunk. ^[142]
G0069	MuddyWater	MuddyWater has used C2 infrastructure to receive exfiltrated data. ^[143]
G0129	Mustang Panda	Mustang Panda has exfiltrated stolen data and files to its C2 server. ^{[144][145][146]}
S0034	NETEAGLE	NETEAGLE is capable of reading files over the C2 channel. ^[15]
S1090	NightClub	NightClub can use SMTP and DNS for file exfiltration and C2. ^[147]
S0385	njRAT	njRAT has used C2 infrastructure to receive stolen information from the infected machine including screenshots and other system information. ^{[148][149]}
S0340	Octopus	Octopus has uploaded stolen files and data from a victim's machine over its C2 channel. ^[150]
S1170	ODAgent	ODAgent can use an attacker-controlled OneDrive account to receive C2 commands and to exfiltrate files. ^[151]
S1172	OilBooster	OilBooster can use an actor-controlled OneDrive account for C2 communication and exfiltration. ^[151]
S0439	Okrum	Data exfiltration is done by Okrum using the already opened channel with the C2 server. ^[152]
S0264	OopsIE	OopsIE can upload files from the victim's machine to its C2 server. ^[153]
C0022	Operation Dream Job	During Operation Dream Job, Lazarus Group exfiltrated data from a compromised host to actor-controlled C2 servers. ^[154]
C0006	Operation Honeybee	During Operation Honeybee, the threat actors uploaded stolen files to their C2 servers. ^[155]
C0014	Operation Wocao	During Operation Wocao, threat actors used the XServer backdoor to exfiltrate data. ^[156]
S1017	OutSteel	OutSteel can upload files from a compromised host over its C2 channel. ^[157]
S1050	PcShare	PcShare can upload files and information from a compromised host to its C2 servers. ^[78]
S0587	Penguin	Penguin can execute the command code <code>do_upload</code> to send files to C2. ^[158]
S9014	PHASEJAM	PHASEJAM has the ability to exfiltrate data from the victim appliance. ^[159]
S1145	Pikabot	During the initial Pikabot command and control check-in, Pikabot will transmit collected system information encrypted using RC4. ^[160]
S1031	PingPull	PingPull has the ability to exfiltrate stolen victim data through its C2 channel. ^[161]
S0013	PlugX	PlugX has exfiltrated stolen data and files to its C2 server. ^{[162][146]}
S0428	PoetRAT	PoetRAT has exfiltrated data over the C2 channel. ^[163]
S1173	PowerExchange	PowerExchange can exfiltrate files via its email C2 channel. ^[164]
S0441	PowerShower	PowerShower has used a PowerShell document stealer module to pack and exfiltrate .txt, .pdf, .xls or .doc files smaller than 5MB that were modified during the past two days. ^[165]

S0238	Proxysvc	Proxysvc performs data exfiltration over the control server channel using a custom protocol. ^[166]
S0078	Psylo	Psylo exfiltrates data to its C2 server over the same protocol as C2 communications. ^[38]
S0147	Pteranodon	Pteranodon exfiltrates screenshot files to its C2 server. ^[80]
S0192	Pupy	Pupy can send screenshots files, keylogger data, files, and recorded audio back to the C2 server. ^[167]
S0650	QakBot	QakBot can send stolen information to C2 nodes including passwords, accounts, and emails. ^[168]
S1148	Raccoon Stealer	Raccoon Stealer uses existing HTTP-based command and control channels for exfiltration. ^{[169][170][171]}
S0495	RDAT	RDAT can exfiltrate data gathered from the infected system via the established Exchange Web Services API C2 channel. ^[172]
S1240	RedLine Stealer	RedLine Stealer has sent victim data to its C2 server or RedLine panel server. ^[173]
C0056	RedPenguin	During RedPenguin, UNC3886 uploaded specified files from compromised devices to a remote server. ^[174]
S0375	Remexi	Remexi performs exfiltration over BITSAdmin, which is also used for the C2 channel. ^[175]
S0496	REvil	REvil can exfiltrate host and malware information to C2 servers. ^[176]
S0448	Rising Sun	Rising Sun can send data gathered from the infected machine via HTTP POST request to the C2. ^[177]
S0240	ROKRAT	ROKRAT can send collected files back over same C2 channel. ^[178]
S1078	RotaJakiro	RotaJakiro sends device and other collected data back to the C2 using the established C2 channels over TCP. ^[179]
S0085	S-Type	S-Type has uploaded data and files from a compromised host to its C2 servers. ^[138]
S1210	Sagerunex	Sagerunex encrypts collected system data then exfiltrates via existing command and control channels. ^[180]
G0034	Sandworm Team	Sandworm Team has sent system information to its C2 server using HTTP. ^[181]
G1015	Scattered Spider	Scattered Spider has exfiltrated data from compromised VMware vCenter servers through an established C2 channel using the Teleport remote access tool. ^[182]
S0461	SDBbot	SDBbot has sent collected data from a compromised host to its C2 servers. ^[76]
S9008	Shai-Hulud	Shai-Hulud has used POST to exfiltrate secrets from the victim environment to an attacker-controlled URL. ^{[183][184][185]}
C0058	SharePoint ToolShell Exploitation	During SharePoint ToolShell Exploitation, threat actors exfiltrated stolen credentials and internal data over HTTPS to C2 infrastructure. ^[186]
S1019	Shark	Shark has the ability to upload files from the compromised host over a DNS or HTTP C2 channel. ^[187]
S1089	SharpDisco	SharpDisco can load a plugin to exfiltrate stolen files to SMB shares also used in C2. ^[147]
S0445	ShimRatReporter	ShimRatReporter sent generated reports to the C2 via HTTP POST requests. ^[188]
S1178	ShrinkLocker	ShrinkLocker will exfiltrate victim system information along with the encryption key via an HTTP POST. ^{[189][190]}
S0610	SideTwist	SideTwist has exfiltrated data over its C2 channel. ^[191]
S0692	SILENTRINITY	SILENTRINITY can transfer files from an infected host to the C2 server. ^[192]
S0633	Sliver	Sliver can exfiltrate files from the victim using the <code>download</code> command. ^[193]
S0533	SLOTHFULMEDIA	SLOTHFULMEDIA has sent system information to a C2 server via HTTP and HTTPS POST requests. ^[194]
S0649	SMOKEDHAM	SMOKEDHAM has exfiltrated data to its C2 server. ^[195]
S1166	Solar	Solar can send staged files to C2 for exfiltration. ^[132]
S0615	SombRAT	SombRAT has uploaded collected data and files from a compromised host to its C2 server. ^[196]
S0543	Spark	Spark has exfiltrated data over the C2 channel. ^[197]
S1030	Squirrelwaffle	Squirrelwaffle has exfiltrated victim data using HTTP POST requests to its C2 servers. ^[198]
S1037	STARWHALE	STARWHALE can exfiltrate collected data to its C2 servers. ^[199]
G0038	Stealth Falcon	After data is collected by Stealth Falcon malware, it is exfiltrated over the existing C2 channel. ^[200]
S1183	StrelaStealer	StrelaStealer exfiltrates collected email credentials via HTTP POST to command and control servers. ^{[201][202][203][204]}
S1034	StrifeWater	StrifeWater can send data and files from a compromised host to its C2 server. ^[205]
S0491	StrongPity	StrongPity can exfiltrate collected documents through C2 channels. ^{[206][207]}
S0603	Stuxnet	Stuxnet sends compromised victim information via HTTP. ^[208]
S1042	SUGARDUMP	SUGARDUMP has sent stolen credentials and other data to its C2 server. ^[209]
S1064	SVCReady	SVCReady can send collected data in JSON format to its C2 server. ^[210]
S0663	SysUpdate	SysUpdate has exfiltrated data over its C2 channel. ^[211]
S0467	TajMahal	TajMahal has the ability to send collected files over its C2. ^[212]
S0595	ThiefQuest	ThiefQuest exfiltrates targeted file extensions in the <code>\\Device\</code> folder to the command and control server via unencrypted HTTP. Network packets contain a string with two pieces of information: a file path and the contents of the file in a base64 encoded string. ^{[213][214]}
S0671	Tomiris	Tomiris can upload files matching a hardcoded set of extensions, such as .doc, .docx, .pdf, and .rar, to its C2 server. ^[215]
S0678	Torisma	Torisma can send victim data to an actor-controlled C2 server. ^[216]
S1201	TRANSLATEXT	TRANSLATEXT has exfiltrated collected credentials to the C2 server. ^[217]
S0266	TrickBot	TrickBot can send information about the compromised host and upload data to a hardcoded C2 server. ^{[218][219]}

S119b	Iroll Stealer	Iroll Stealer exfiltrates collected information to its command and control infrastructure. ^{[64][65]}
S0386	Ursnif	Ursnif has used HTTP POSTs to exfil gathered information. ^{[221][222][223]}
S0476	Valak	Valak has the ability to exfiltrate data over the C2 channel. ^{[224][225][226]}
G1055	VOID MANTICORE	VOID MANTICORE malware has exfiltrated collected data via Telegram bot C2 channels using encrypted communications. ^[227]
S0670	WarzoneRAT	WarzoneRAT can send collected victim data to its C2 server. ^[228]
G1035	Winter Vivern	Winter Vivern delivered a PowerShell script capable of recursively scanning victim machines looking for various file types before exfiltrating identified files via HTTP. ^[229]
G0090	WIRTE	WIRTE has exfiltrated collected victim data to C2 infrastructure. ^[11]
G0102	Wizard Spider	Wizard Spider has exfiltrated domain credentials and network enumeration information over command and control (C2) channels. ^{[230][231]}
S1065	Woody RAT	Woody RAT can exfiltrate files from an infected machine to its C2 server. ^[232]
S0658	XCSSET	XCSSET retrieves files that match the pattern defined in the INAME_QUERY variable within the user's home directory, such as <code>*teat.txt</code> , and are below a specific size limit. It then archives the files and exfiltrates the data over its C2 channel. ^{[233][234]}
S1248	XORIndex Loader	XORIndex Loader has exfiltrated victim data using HTTPS POST requests to its C2 servers. ^[20]
S0251	Zebrocy	Zebrocy has exfiltrated data to the designated C2 server using HTTP POST requests. ^{[235][236]}
G0128	ZIRCONIUM	ZIRCONIUM has exfiltrated files via the Dropbox API C2. ^[237]
S0086	ZLib	ZLib has sent data and files from a compromised host to its C2 servers. ^[138]

Mitigations

ID	Mitigation	Description
M1057	Data Loss Prevention	Data loss prevention can detect and block sensitive data being sent over unencrypted protocols.
M1031	Network Intrusion Prevention	Network intrusion detection and prevention systems that use network signatures to identify traffic for specific adversary malware can be used to mitigate activity at the network level. Signatures are often for unique indicators within protocols and may be based on the specific obfuscation technique used by a particular adversary or tool, and will likely be different across various malware families and versions. Adversaries will likely change tool command and control signatures over time or construct protocols in such a way to avoid detection by common defensive tools. ^[238]

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0348	Detection Strategy for Exfiltration Over C2 Channel	AN0988	Identifies suspicious outbound traffic volume mismatches from processes that typically do not generate network activity, particularly over C2 protocols like HTTPS, DNS, or custom TCP/UDP ports, following file or data access.
		AN0989	Monitors for processes reading sensitive files then immediately initiating unusual outbound connections or bulk transfer sessions over persistent sockets, particularly with encrypted or binary payloads.
		AN0990	Detects unauthorized applications or scripts accessing sensitive data followed by establishing encrypted outbound communication to rare external destinations or with abnormal byte ratios.
		AN0991	Detects VMs sending outbound traffic through non-standard services or to unknown destinations. Exfiltration over reverse shells tunneled via VMkernel or custom payloads routed via hostd/vpxa.

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